

Metagenomic and Bioinformatic Tools for Deep-Ocean Observation and Diversity Studies

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Main Goal

Providing an **overview and practical application** of current methodologies, such as **eDNA metabarcoding, metagenomics**, and associated bioinformatic tools to analyze next-generation sequencing data and to make insightful interpretations in the context of marine ecology and biodiversity (**with a focus on microbiology**)

- Overview of different tools, not super in-depth for a single tool
- Metabarcoding (targeted) and Metagenomics (untargeted)
- 50/50 theory and practical (two days theoretical-practical, two days hands-on)
- Note for future work...**scripts here are not immediately applicable**. Critical thinking and time-spent are key for successful and meaningful work. Take your time to develop, think, and explore your own analysis workflow.
- Applicable to a wide range of research questions (even though intrinsically biased)
- Motivation: increasing need from multiple environmental and marine observation fields

Practical Info

- WIFI network: **eduroam** (username: sc.wifi.4@fc.up.pt; password: FCUP05Maio)
- Coffee-breaks provided at FC4 (main stairs near main entrance) and lunch at the Multirest restaurant (building next door)
- Posters can be displayed in the coffee break area and will be presented/discussed in the afternoon coffee breaks
- Github link with materials (presentations, scripts, data – **email with link will be sent at lunchtime**): <https://github.com/Nik3939/Metagenomic-and-bioinformatic-tools-for-deep-ocean-observation-and-diversity-studies>
- Metabarcoding hands-on session (Day 3 and Day 4): **Who has R and Rstudio installed? And packages?**
- Metagenomics hands-on session (Day 3 and Day 4): instructions to connect to server will be provided

Program (Days 1 and 2) - TP

Lecture and presentation type of sessions (same room). With a few brief examples.

Metabarcoding

Day 1 (05/05/2026) Theoretical-practical: Metabarcoding		
08h45-09h15	Welcome and intro to the workshop (JX or IG, MS)	FC5.003
09h15-10h15	Keynote Alessandro Vezzi - Keeping it short, going long, and trusting Sanger: From Sanger sequencing to NGS	FC5.003
10h15-10h30	Coffee Break	
10h30-11h00	eDNA sampling protocols and standardization for microbiome studies (CM)	FC5.003
11h00-12h00	The microbial realm and metabarcoding/amplicon sequencing for diversity studies (MS)	FC5.003
12h00-13h15	Lunch	
13h15-14h15	Practical demonstration: Silva NGS (CM)	FC4.011
14h15-15h00	R tools for Microbial Ecology (FP)	FC4.011
15h00-15h30	Coffee Break (with posters)	
15h30-17h00	Metabarcoding analysis workflow in R (MS and FP)	FC4.011

Metagenomics

Day 2 (06/05/2026) Theoretical-practical: Metagenomics		
09h00-09h15	Plan for the day	FC5.003
09h15-10h15	Keynote Ana Riesgo - Fantastic natural samplers and where to find them	FC5.003
10h15-10:30	Coffee Break	
10h30-12h00	Metagenomics: From reads to MAGs (NG)	FC5.003
12h00-13h15	Lunch	
13h15-14h15	Taxonomic and Functional annotation of MAGs (AR)	FC4.011
14h15-15h00	Introduction to Python programming (NG)	FC4.011
15h00-15h30	Coffee Break (with posters)	
15h30-16h15	eDNA from Gorrige TwinDEEPS expedition (AC and JC)	FC4.011
16:15-17:00h	Transcriptomics in non-model organisms and insights into metatranscriptomics (AV)	FC4.011

Program (Days 3 and 4) - P

Two parallel hands-on sessions (FC4.011 Metabarcoding and FC4.012 Metagenomics)

Day 3 (07/05/2026) “Hands-on” Sessions		
09h00-09h15	Plan for the day Room FC4.011	
09h15-10h15	Exploring online resources and repositories (MS) Room FC4.011	
10h15-10h30	Coffee Break – split in two rooms	
10h30-12h00	FC4.011 (Metabarcoding) From raw 16S data to abundance and diversity: Metal exposure experiments with deep-sea sediments. <i>Dada2 in R</i> (MS)	FC4.012 (Metagenomics) Introduction to terminal & Quality control of NGS data (AR & NG)
12h00-13h15	Lunch	
13h15-15h00	Data preparation and diversity metrics (FP)	Metagenomic assembly (AR & NG)
15h00-15h30	Coffee Break (with posters)	
15h30-17h00	Community composition and taxonomy (FP)	Metagenomic binning (AR & NG)

Day 4 (08/05/2026) “Hands-on” Sessions (cont.)		
09h00-10h15	FC4.011 (Metabarcoding) Questions and discussion, and continuation of previous work (MS & FP)	FC4.012 (Metagenomics). Taxonomic assignment & Functional prediction (Primary metabolism) (AR & NG)
10h15-10h30	Coffee Break	
10h30-12h00	Exploring heterogeneity: when variance <i>beats</i> mean (MS)	Functional prediction (Secondary metabolism) (AR & NG)
12h00-13h15	Lunch (rejoin same room for presentations)	
13h15-15h00	Group discussion and exercise (rooms available: FC5.003 and FC4.011)	
15h00-15h30	Coffee Break	
15h30-17h00	Group presentations (FC5.003)	

Show of hands:

- 1) Metabarcoding:
- 2) Metagenomics:
- 3) Don't know:
- 4) Self- cloning and joining both:

6 Instructors

Metabarcoding



Catarina Magalhães

Associate Professor (FCUP)
Group Leader (CIIMAR)



Miguel Semedo

Junior Researcher (CIIMAR)



Francisco Pascoal

Junior Researcher (CIIMAR)

Metagenomics



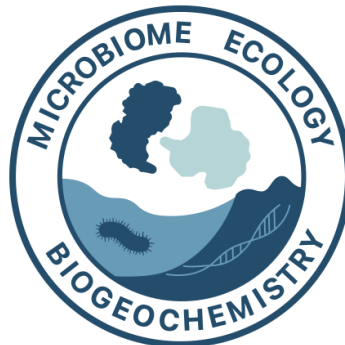
Adriana Rego

Junior Researcher
(CIIMAR)



Nicola Gambardella

PhD Student (CIIMAR)



Keynote and Invited Instructor



Alessandro Vezzi

Senior Researcher

University of
Padova
(Department of
Biology)

28 participants (mostly from CIIMAR)

Stage		
MSc Student	6	23%
PhD Student	5	19%
Research Technician	2	8%
Junior Researcher/Postdoc	8	31%
Assistant Researcher	1	4%
Other	4	15%

Experience in high-throughput technologies		
1- No experience	7	27%
2 - Beginner	7	27%
3 - Intermediate	7	27%
4 - Experienced	5	19%
5 - Advanced	0	0%

Experience in R		
1- No experience	4	15%
2 - Beginner	10	38%
3 - Intermediate	8	31%
4 - Experienced	4	15%
5 - Advanced	0	0%

Experience in Python		
1- No experience	13	50%
2 - Beginner	11	42%
3 - Intermediate	0	0%
4 - Experienced	2	8%
5 - Advanced	0	0%

- Name
- Affiliation + lab/group
- Current position (MSc Student/PhD student/Postdoc, etc.)
- Research area/main topic of research